DINet: Depth-Guided and Iterative Refinement Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*—Optical remote sensing images (ORSIs) feature unique scenes and complex imaging conditions. Specifically, they exhibit substantial variations in object scale, quantity, structure, and distribution. Consequently, salient object detection (SOD) in ORSI (ORSI-SOD) is pivotal in ORSI content perception and understanding. In addition, the limitations of the single modality impede the advancement of ORSI-SOD. To tackle these issues, we propose a depth-guided and iterative refinement network (DINet) for ORSI-SOD. By incorporating depth information as auxiliary cues, we introduce a multimodal strategy for ORSI-SOD, resulting in improved accuracy in the localization and segmentation of salient objects. To address the variability of salient objects, we design an aggregation perception enhancement (APE) module. This module integrates complementary cues from cross-modal features using multidimensional attention mechanisms. By fostering cross-modal interactions, the APE module effectively preserves both detail and spatial location information. Furthermore, we propose an iterative guidance refinement decoder to handle boundary uncertainty. The decoder uses initial predictions to guide the decoding phase and iteratively refine results. Simultaneously, it minimizes noise from depth cues, yielding predictions with more accurate boundaries. Experimental comparisons with 22 state-of-the-art methods show that DINet exhibits superior performance while maintaining lightweight (11.98 M) and real-time (55 FPS) capabilities.**

***Index Terms*—Cross-modality, iterative refinement, optical remote sensing images (ORSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) is designed to locate and segment the most prominent objects or regions in a scene by simulating human visual attention mechanisms. By effectively extracting valuable information from images, SOD facilitates further processing and has a wide range of applications, including image segmentation [1], image fusion [2], and object tracking [3].

Research on SOD has emerged and primarily focused on natural scene images (NSIs), and a series of studies have emerged [4], [5], [6], [7], which establish a foundation for SOD in specific scenarios. Recently, the potential of optical remote sensing images (ORSI) for broad applications in fields such as agriculture [8] and military [9] has made ORSI-SOD a new research focus. The spectral bands of ORSI obtained from airborne sensors and satellites predominantly range from 400 to 700 nm, with objects mainly including airplanes, buildings, ships, rivers, and roads. Overall, ORSI differs significantly from NSI in object scale, quantity, contours, and field of view. Consequently, directly applying existing NSI-SOD methods to ORSI-SOD is often inadequate.

As illustrated in Fig. 1, ORSI-SOD faces several inherent challenges. First, scale disparity is a major issue due to the large field of view. Small objects like airplanes occupy a minimal proportion, whereas large objects like rivers cover substantial areas. This significant scale disparity increases the difficulty of accurately locating salient objects. In addition, issues such as insufficient illumination and shadow interference further complicate the accurate localization of objects. Second, scenes featuring multiple small objects or lacking salient objects often result in missed or incorrectly detected objects. Finally, the irregular and diverse contours of objects in ORSI present challenges for detecting precise boundaries.

Addressing the aforementioned issues, incorporating multimodal information, such as depth maps, holds promise in providing sufficient discriminative information. In NSI, salient objects are typically close to the camera and occupy a compact area, so the depth map, which includes information such as perspective and actual distance, can provide a strong salient signal. However, due to ORSI's large field of view, the foreground and background may be at almost the same height, making it difficult to rely directly on absolute depth in a metric sense. Despite these differences, depth maps provide valuable spatial layout and structural protrusion information, which can effectively enhance the network's localization capability in complex environments. This makes depth guidance particularly suitable for ORSI-SOD, where objects often exhibit significant scale variations and are embedded in cluttered backgrounds. Moreover, the integration of depth cues helps

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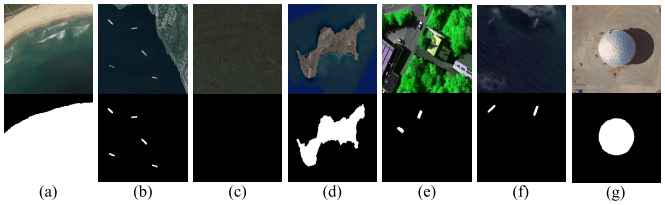


Fig. 1. Representative examples of remote sensing objects. (a) Large object. (b) Multiple small objects. (c) No salient object. (d) Complex boundary. (e) Cluttered background. (f) Insufficient lighting. (g) Shadow interference.

to disambiguate objects from backgrounds with similar color or texture, a common challenge in ORSI.

Compared to RGB images, depth maps contain more spatial positioning information, specifically manifested as the separation of structural protrusions from planar regions, relative height information, and scale and texture information of ground and objects. This effectively enhances the network's localization capability in complex environments. However, RGB-D SOD research specifically for ORSI has been scarcely explored. Due to the high acquisition difficulty and cost of ORSI, no public RGB-D datasets are currently available, which limits the development of multimodal ORSI-SOD research. Therefore, using depth maps generated by monocular depth estimation (MDE) is a feasible approach to conducting multimodal ORSI-SOD research. However, depth maps generated via MDE can be unreliable. These maps inevitably introduce noise or negative information, posing significant challenges for precise boundary segmentation in ORSI-SOD.

To address these challenges, we explore the potential of existing ORSI from the perspective of multimodal SOD. By utilizing MDE to generate depth images from existing ORSI datasets as supplementary cues to guide RGB images, we design a novel ORSI-SOD method, depth-guided and iterative refinement network (DINet), which employs a multimodal strategy and achieves state-of-the-art performance. This approach ensures accurate localization and segmentation of salient objects in ORSI through three key processes: shared weight feature extraction, complementary information aggregation enhancement, and iterative guidance refinement decoding.

Specifically, during feature extraction, we design a shared weight feature extraction encoder, differing from the commonly used dual-stream backbones in most RGB-D SOD methods, due to the smaller distribution differences between generated depth maps and RGB maps. This effectively reduces the number of parameters while promoting interaction between cross-modal features. In addition, to address the extra computational burden of incorporating depth cues, we adopt a hybrid architecture backbone with lower computational costs, enabling the effective capture of global dependencies while preserving detailed information. In the feature aggregation phase, we propose an aggregation perception enhancement (APE) module. It aggregates complementary cross-modal information to enhance RGB detail representation while retaining spatial cues from depth images. This approach improves the network's localization ability for objects with significant scale variations. During the decoding and prediction phase, we propose an iterative guidance refinement decoder that integrates multiscale contextual information, using semantic information from high-level features to guide the process and reduce interference, thus yielding more accurate predictions. Furthermore, the initial prediction is iterated as an attention map, allowing the network to refine salient predictions further and mitigate the negative noise introduced by the generated depth maps, resulting in clearer and more accurate boundaries for the salient predictions.

In summary, the main contributions of this article are as follows.

1) We explore and leverage the potential of existing ORSIs from the perspective of multimodal SOD. By utilizing depth cues as guidance, we develop a multimodal ORSI-SOD model, DINet, aimed at promoting research in multimodal ORSI-SOD.
2) We design an APE module that employs attention mechanisms across multiple dimensions to perceive and aggregate complementary cues in RGB and depth features. This enables the complementary fusion of cross-modal features while enhancing the representation of detailed and locational information.
3) We propose an iterative guidance refinement decoder that combines multiscale contextual information to address uncertainties associated with the scale of remote sensing objects. In addition, through an iterative guidance strategy that progressively refines difficult boundary regions, DINet addresses the irregular contours of objects in ORSI.
4) Results on three ORSI-SOD datasets demonstrate that DINet effectively confronts the inherent challenges of ORSI-SOD and achieves significant performance improvements compared to state-of-the-art networks, validating the potential of multimodal ORSI-SOD models.

II. Related Work

A. SOD for NSI and ORSI

SOD is an upstream computer vision task. Early studies primarily centered on traditional image processing methods, detecting salient objects by introducing heuristic priors and handcrafted features [4], [5]. These methods laid the foundation for the SOD research. With the rise of deep learning, convolutional neural network (CNN)-based methods for SOD have made significant breakthroughs, resulting in a proliferation of related studies [6], [7]. Recently, Transformer has gained attention as a key area of research in computer vision [10], [11], which has greatly promoted the study of SOD. Liu et al. [12] proposed a novel architecture entirely on the Transformer, effectively adapting to dense prediction tasks through multilevel token fusion. Ren et al. [13] devised a global-local Transformer architecture that achieved significant performance enhancements. Benefits to the powerful long-range dependency modeling capability of Transformer, the global performance of these methods has been greatly improved.

The emergence of publicly available ORSI datasets [14], [15], [16] has promoted the expansion of SOD related research

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from NSI to ORSI. Early research attempted to apply NSI-SOD methods directly to ORSI-SOD. However, direct transfer often yields suboptimal results due to significant differences in object scale, background complexity, and imaging conditions. Consequently, researchers have embarked on a series of studies focusing on the characteristics of ORSI. For instance, to cope with complex background interference, Quan et al. [17] proposed SFANet, introducing an uncertainty-aware mechanism to suppress background noise. To tackle multiscale object detection challenges, Yan et al. [18] designed a Transformer-CNN dual-branch encoder that balances global semantics and local details. To mitigate noise caused by feature redundancy, Di et al. [19] developed a two-stage network that enhances feature purity through multiscale weighting and indirect fusion. Furthermore, to improve model practicality and deployment efficiency, Li et al. [20] proposed a lightweight network based on semantic matching and edge alignment, reducing computational load while maintaining performance. Despite the aforementioned methods having achieved certain progress, existing ORSI-SOD still face fundamental challenges, including significant scale variations of salient objects, an uncertain number of objects, interference from lighting variations and shadows, as well as imaging blur. These factors lead to notable deficiencies in existing models regarding localization accuracy, boundary clarity, and robustness in complex scenarios. Consequently, relying solely on RGB modal information can hardly lead to further breakthroughs. Introducing additional discriminative cues, such as depth information, has become a critical direction for enhancing ORSI-SOD performance.

B. Monocular Depth Estimation

Existing RGB-D datasets primarily focus on natural scenarios. Due to the high costs associated with obtaining ORSI data, there are currently no publicly available RGB-D datasets. MDE aims to predict the depth of each pixel in an input RGB image. MDE methods are generally categorized into discriminative and generative models based on their architectural approaches. Generative model-based methods tend to provide finer detail modeling, while discriminative model-based methods offer superior robustness and lightweight characteristics. For instance, Marigold [21] achieved impressive MDE by fine-tuning a latent diffusion model. Zoe [22] proposed the first method that effectively combines relative depth estimation with metric depth estimation, significantly enhancing generalization performance. Depth Anything (DA) [23] excelled in implementing zero-shot depth estimation using a large volume of unlabeled data. DA V2 (DA2) [24] further optimized this approach, greatly improving model efficiency and generating finer, more robust predictions, thus successfully establishing a universal foundational model for depth estimation. Building on this research, we can leverage MDE technology to acquire depth maps for ORSI cost-effectively and efficiently, enabling the development of multimodal ORSI-SOD.

C. RGB-D Salient Object Detection

In NSI-SOD research, RGB-D methods have effectively enhanced models' perception and discrimination capabilities in complex scenes by incorporating spatial structure and layout information provided by depth maps, thereby overcoming the performance limitations of single-modal RGB approaches. Yao et al. [25] proposed a depth injection strategy that efficiently utilizes depth information to enhance RGB feature representation while maintaining computational efficiency. CPNet [26] addressed feature redundancy and performance degradation caused by additional fusion modules through a simplified architectural design. To tackle the challenge of unreliable depth map quality, Zhao et al. [27] introduced a depth-aware framework that uses depth data as a supervisory signal for representation learning without requiring depth input during testing.

However, these methods are all designed for NSI, based on the premise that object distance in depth maps is highly correlated with saliency. In ORSI, due to the top-down perspective and wide field of view, foreground and background are often at similar heights, diminishing the discriminative significance of absolute physical depth. Instead, the primary value of depth maps in ORSI lies in providing spatial layout, structural convexity, and relative hierarchical relationships. Furthermore, due to the current lack of real RGB-D datasets for ORSI, this article employs MDE methods to generate depth maps. Such pseudo-depth maps inevitably contain noise and estimation inaccuracies. Directly applying fusion strategies designed for NSI may introduce significant interference. Therefore, this article does not treat depth maps as an equivalent modality to RGB but rather as auxiliary cues to enhance the model's spatial structure perception and object localization capabilities in complex remote sensing scenarios.

III. Proposed Method

A. Overview Architecture

The overall architecture of the proposed multimodal ORSI-SOD method, DINet, is shown in Fig. 2, following an encoder-fusion-decoder design. First, the RGB image is preprocessed with DA2 [24] to generate the depth image. Next, during the encoder phase, features are extracted from RGB images and generated depth images using a shared weight Transformer backbone, yielding hierarchical features that encompass both global and detailed information. Then, an APE module integrates spatial location and detailed information from the multimodal features through attention perception across multiple dimensions. Finally, an iterative guidance refinement decoder consists of iterative guidance units (IGUs) and residual attention units (RAUs) that decode the cross-modal fused features, iteratively refining the salient prediction results through multistage cycles while supervising predictions at each stage, ultimately producing clear and accurate predictions. Detailed information regarding each component is provided below.

B. Shared Weight Encoder

The salient objects in ORSI exhibit significant size differences and irregular contours. Therefore, effectively utilizing global and local information at various scales is crucial for successful detection. While CNN-based feature encoders

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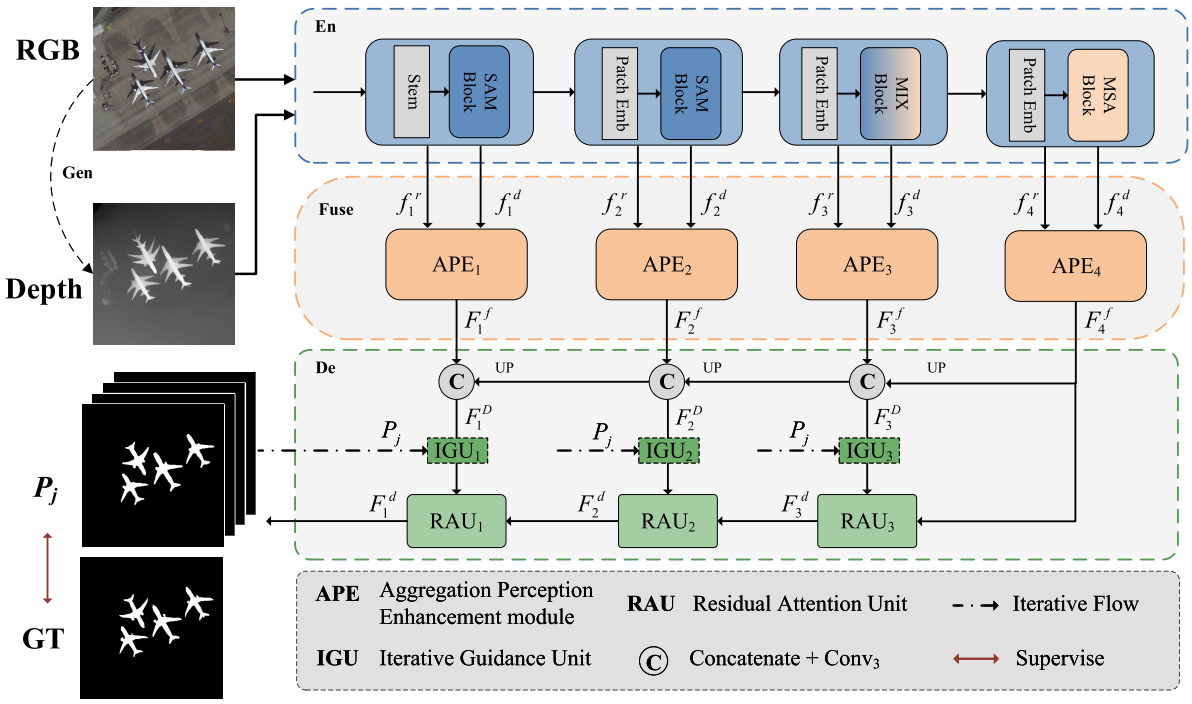


Fig. 2. Overall framework of DINet. For the RGB map and the depth map generated from it, multilevel predictions are obtained through three stages: a shared-weight encoder, cross-modal complementary fusion, and an iterative guidance refinement decoder, with multilevel supervision applied via a hybrid loss function.

generally excel in modeling local details, their limitations in local receptive fields can result in incomplete detection or missed objects. In contrast, the self-attention mechanism of the Transformer offers powerful global modeling capabilities, allowing for the effective capture of long-range dependencies, thus addressing this challenge. Combining CNNs and Transformers can leverage the strengths of both, enabling comprehensive modeling of global and local information while reducing computational burdens.

To this end, we employ a hybrid architecture known as the scale-aware modulation transformer (SMT) [10] as the encoder for feature extraction. SMT is structured into four stages. In the first two stages, which utilize a CNN-based approach, scale-aware modulation (SAM) is implemented to extract multiscale local detail information from high-resolution input features while reducing feature resolution. This significantly alleviates the computational load that would arise from applying the self-attention mechanism directly to high-resolution images. In the third stage, a MIX block of stacked CNN and Transformer layers facilitates the transition from local detail to global semantic information. The final stage incorporates a pure Transformer structure to enhance the acquisition of global semantic information. This hierarchical mixed architecture allows SMT to effectively capture multiscale hierarchical features that include both local details and global semantics for further processing. Moreover, the depth features in this study are generated directly from RGB features. Therefore, the distribution gap between these generated depth maps and RGB images is smaller than that of sensor-captured depth maps. This implies they share more common information. For this reason, we employ a shared weight encoder rather than the dual-stream encoders typically used in most RGB-D SOD methods, effectively reducing the model's parameter count and enabling initial interaction between cross-modal features. Specifically, depth images are replicated into three channels to align with the RGB images. RGB features and depth features are then simultaneously input into the SMT encoder to obtain multiscale features $f_i^r(i = 1, 2, 3, 4)$ and $f_i^d(i = 1, 2, 3, 4)$. To ensure a lightweight network, we utilize the SMT-T version with the minimum parameter count.

C. APE Module

To address the localization challenges in ORSI caused by extreme scale variations of salient objects and complex spatial distributions, we designed the APE module. The core of this module lies in its oriented perception and cross-interaction fusion strategy tailored for ORSI, which explicitly guides the complementary integration of detailed information from the RGB modality and spatial structural information from the depth modality. Through a multidimensional, grouped cooperative attention mechanism, it effectively perceives and aggregates complementary cues across modalities.

The structure of the APE module is illustrated in Fig. 3. For the multiscale cross-modal features $f_i^{r/d}(i = 1, 2, 3, 4)$ extracted from the encoder, global spatial information is first compressed through two 1-D average pooling operations $P_{x/y}(\cdot)$, while retaining position information in the 1-D direction. The 1-D direction perception features from both modalities are then cross-concatenated to produce P_{xy} and P_{yx}, with initial interactive combined cross-modal features f_p derived through multiplication and addition operations

$$P_{xy} = \mathrm{Con}\left(P_x\left(f_i^r\right), P_y\left(f_i^d\right)\right) \tag{1}$$

$$P_{yx} = \mathrm{Con}\left(P_y\left(f_i^r\right), P_x\left(f_i^d\right)\right) \tag{2}$$

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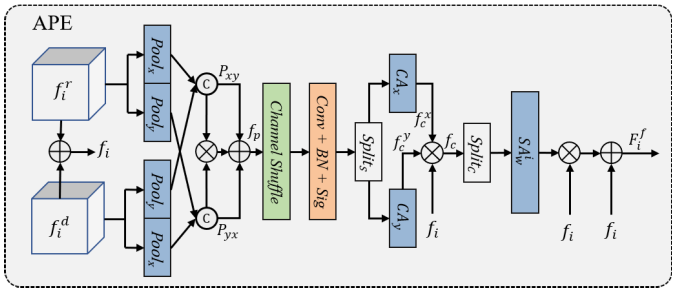


Fig. 3. Detailed architecture of the proposed APE module.

$$f_p = P_{xy} + P_{yx} + \left(P_{xy} \cdot P_{xy}\right) \tag{3}$$

where Con(·) represents the channel concatenation operation. The channel shuffle operation CS(·) is then employed to achieve cross-channel information interaction, followed by a convolution layer equipped with BatchNorm and Sigmoid ConvBS(·) to integrate and compress channel information. Subsequently, the spatial features embedded with directional information are split along the 1-D spatial direction through $\text{Split}_s(\cdot)$. A coordinate attention [28] layer $CA_{x/y}(\cdot)$ generates 1-D coordinate attention features f_c^x and f_c^y, which are multiplied with the input features $f_i = f_i^r + f_i^d$ to achieve attention perception in the channel dimension. This can be described as

$$f_c^{x/y} = CA_{x/y}\left(\text{Split}_s\left(\text{ConvBS}\left(\text{CS}\left(f_p\right)\right)\right)\right) \tag{4}$$

$$f_c = f_i \times f_c^x \times f_c^y. \tag{5}$$

Through these steps, global information is encoded, enabling attention perception in the channel dimension while preserving location information.

Next, the channel attention perception features f_c are divided into four groups along the channel dimension using $\text{Split}_c(\cdot)$. Spatial attention branch features f_s^i are obtained by employing a spatial attention module $\text{SA}_w^i(\cdot)$ with learnable weight parameters. These branch features are then aggregated to form the spatial attention perception features f_s. Finally, these features are multiplied by the input features f_i and combined using residual calculations to yield the output attention perception fusion features F_i^f. This process is described as

$$f_s^i = \text{SA}_\omega^i\left(\text{Split}_c\left(f_c\right)\right), i = 1, 2, 3, 4 \tag{6}$$

$$f_s = \sum_{i=1}^{4} f_s^i \tag{7}$$

$$F_i^f = f_s \times f_i + f_i. \tag{8}$$

By adopting this grouped spatial attention approach, we can preserve a wider variety of information. The inclusion of learnable parameters further enhances the expressive capability of this differential information.

D. Iterative Guidance Refinement Decoder

A significant challenge in ORSI-SOD is the irregular and diverse contours of salient objects, which complicates the task of obtaining clear and complete object boundaries. To tackle this issue, we design an iterative guidance refinement decoder that integrates multiscale information from contextual features to enrich the predicted details. Simultaneously, the initial predictions undergo multiple rounds of iterative refinement, where IGUs further refine the features. Ultimately, RAUs are employed to enhance feature perception and integration, yielding clear and comprehensive salient predictions.

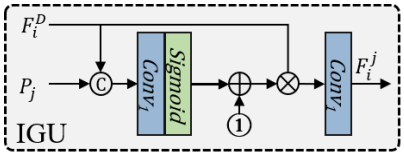


Fig. 4. Illustration of the IGU.

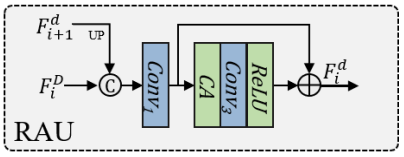


Fig. 5. Illustration of the RAU.

As shown in Fig. 2, the iterative guidance refinement decoder consists of two stages. In the first stage, low-level features are guided by the semantic information derived from high-level features, initially integrating multiscale information from contextual features while reducing the negative impact of noise in low-level features. The specific process is as follows:

$$F_i^D = \text{Conv}_3\left(\text{Con}\left(F_i^f, \text{UP}\left(F_{i+1}^f\right)\right)\right), i = 1, 2, 3 \tag{9}$$

where UP(·) denotes bilinear up-sampling and $\text{Conv}_3(\cdot)$ refers to a 3×3 convolution layer. After obtaining the initial predictions P_1, the stage predictions P_j are processed iteratively. We employ a gate attention-based IGU to guide the integrated features, thereby enhancing the representation capability of salient features

$$F_i^j = \text{IGU}_i\left(P_j, F_i^D\right), i, j = 1, 2, 3 \tag{10}$$

where IGU(·) represents the IGU, with its internal structure illustrated in Fig. 4. The input features F_i^D and the stage predictions P_j are concatenated, followed by channel compression via a 1×1 convolution layer $\text{Conv}_1(\cdot)$. A gate attention map is then generated by Sig(·) and multiplied with the input features F_i^D for attention perception. Finally, the output dimension is adjusted by $\text{Conv}_1(\cdot)$

$$F_i^j = \text{Conv}_1\left(F_i^D \times \left(1 + \text{Sig}\left(\text{Conv}_1\left(\text{Con}\left(F_i^D, P_j\right)\right)\right)\right)\right). \tag{11}$$

In the second stage, the RAU is employed to reintegrate contextual features, filtering out negative information and reinforcing the expression capability of the features to yield salient prediction results. The specific process is as follows:

$$F_i^d = \begin{cases} \text{RAU}_i\left(F_i^D, \text{UP}\left(F_i^d\right)\right), & i = 1, 2 \\ \text{RAU}_i\left(F_i^D, \text{UP}\left(F_{i+1}^f\right)\right), & i = 3 \end{cases} \tag{12}$$

where RAU(·) denotes the RAU, with its internal structure depicted in Fig. 5. The features F_{i+1}^d are up-sampled and concatenated with F_i^d, followed by a channel adjustment

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$\text{Conv}_1(\cdot)$. The features then proceed through channel attention blocks $\text{CA}(\cdot)$, convolution layers $\text{Conv}_3(\cdot)$, and $\text{ReLU}(\cdot)$ to obtain the channel attention map. Finally, channel attention perception is achieved through residual connections to produce the output features F_i^d

$$f = \text{Conv}_1\left(\text{Con}\left(F_{i+1}^d, F_i^D\right)\right) \tag{13}$$

$$F_i^d = f + \text{ReLU}\left(\text{Conv}_3\left(\text{CA}\left(f\right)\right)\right). \tag{14}$$

Note that after the initial iteration, replace F_i^D with F_i^j as the input for $\text{RAU}(\cdot)$.

E. Loss Function

This article employs a multilevel hybrid loss function to supervise the multilevel salient predictions. Specifically, for the initial rounds of the decoder and the salient prediction maps generated during three rounds of iteration, a hybrid loss composed of binary cross-entropy (BCE) loss and intersection over union (IoU) loss is utilized for supervision. This hybrid loss function facilitates more comprehensive supervision over pixel-level prediction details and foreground-background area information. Moreover, the multilevel supervision strategy effectively utilizes different levels of refined features, enhancing the network's discriminative capability while retaining more detailed information.

The BCE loss L_{BCE} is defined as

$$L_{\text{BCE}} = -\sum_{x=1}^{H}\sum_{y=1}^{W}\left[G\left(x,y\right)\log\left(P\left(x,y\right)\right) + \left(1-G\left(x,y\right)\right)\log\left(1-P\left(x,y\right)\right)\right]. \tag{15}$$

The IoU loss L_{IoU} is defined as

$$L_{\text{IoU}} = 1 - \frac{\sum_{x=1}^{H}\sum_{y=1}^{W}P\left(x,y\right)G\left(x,y\right)}{\sum_{x=1}^{H}\sum_{y=1}^{W}\left[P\left(x,y\right)+G\left(x,y\right)-P\left(x,y\right)G\left(x,y\right)\right]}. \tag{16}$$

The overall loss L of the model is defined as

$$L = \sum_{i=1}^{4} L_{\text{BCE}}^i\left(P^i, G\right) + L_{\text{IoU}}^i\left(P^i, G\right) \tag{17}$$

where P^i is the prediction map and G is the ground truth map.

IV. Experiments

A. Experimental Setup

1) Datasets: To evaluate DINet, we conducted tests on three benchmark ORSI datasets: ORSI-4199 [14], EORSSD [15], and ORSSD [16]. Specifically, ORSI-4199 consists of 4199 image pairs, with 2000 images allocated to the training set and 2199 images to the test set. The EORSSD dataset comprises 2000 image pairs, with the training set containing 1400 images and the test set containing 600 images. The ORSSD dataset consists of 800 image pairs, with 600 images in the training set and 200 images in the test set.

2) Evaluation Metrics: We adopted several evaluation metrics to assess the performance of DINet and the comparison methods, including S-measure, F-measure, E-measure, and mean absolute error (MAE). For both F-measure and E-measure, we considered their mean (m) and adaptive (adp) values.

3) Implementation Details: During the training and testing phases, input images are resized to 384×384. Data augmentation techniques such as random flipping, rotation, and boundary cropping are employed during training to prevent overfitting. We utilize the SMT-T pretrained model to initialize the backbone network parameters, while the remaining parameters are set according to PyTorch's default configurations. We train our network using the Adam optimizer, with a batch size of 8 and an initial learning rate of $5e^{-5}$, which is decreased by a factor of 10 every 100 epochs. Our model is trained for approximately 150 epochs on a machine equipped with a single NVIDIA GTX 4090 GPU; the inference speed is about 55 FPS.

B. Comparison With State-of-the-Art Methods

To evaluate the performance of the proposed DINet, we performed qualitative and quantitative comparisons with six NSI-SOD methods and 16 ORSI-SOD methods across the ORSI-4199, EORSSD, and ORSSD datasets. The NSI-SOD methods include DPORTNet [29], DNTD [30], ICON [31], RGB-D methods PICR-Net [32], LAFB [33], and HENet [34]. The ORSI-SOD methods comprise MCCNet [35], HFANet [36], SeaNet [20], ACCoNet [37], GeleNet [38], SFANet [17], MCP-Net [39], ASNet [18], WeightNet [19], LiteSalNet [40], LGIPNet [41], RAMENet [42], MRBINet [43], SggNet [44], BCAR-Net [45], and DPU-Former [46]. The prediction maps for these methods were obtained from the authors and public benchmarks or generated by running publicly available source code; a few of the unobtainable results are indicated with "-."

1) Quantitative Comparison: The quantitative comparison results of DINet against the other 22 methods are presented in Table I. The result shows that DINet achieved the best results in 17 out of 18 metrics, and its overall performance is significantly superior to the existing SOTA methods. Moreover, DINet exhibits substantial improvements over the SOTA methods in many metrics. Specifically, take the EORSSD dataset as an example, compared to the DPU-Former [46] method, DINet achieves improvements of 0.29%, 1.03%, 3.78%, 0.42%, 1.50%, and 0.13 in metrics S_α, F_β^m, F_β^{adp}, E_ξ^m, E_ξ^{adp}, and $\mathcal{M}$, respectively. Additional enhancements were also evident in other datasets. The above quantitative results intuitively demonstrate that our DINet is more robust and competitive.

2) Qualitative Comparison: Fig. 6 displays the qualitative comparison results of DINet with eight SOTA methods in various representative ORSI scenes. DINet utilizes multimodal cues and exceptional iterative refinement capabilities. Consequently, it achieves remarkable detection and segmentation results in challenging scenarios, such as those with multiple objects, complex backgrounds, or intricate edges. Specifically, compared to other methods, in scenes with multiple small objects in Rows 1 and 7, DINet successfully detects all salient

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TABLE I

QUANTITATIVE COMPARISONS WITH NSI-SOD AND ORSI-SOD METHODS ON THREE DATASETS. **RED** AND **BLUE** DENOTE THE BEST AND THE SECOND-BEST RESULTS, RESPECTIVELY. "↑" AND "↓" INDICATE THAT LARGER OR SMALLER IS BETTER. "-" INDICATES THE RESULTS ARE NOT AVAILABLE

Methods	EORSSD						ORSSD						ORSI-4199					
	$S_\alpha\uparrow$	$F_\beta^m\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^m\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^m\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^m\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^m\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^m\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
NSI SOD Methods																		
DPORTNet$_{22}$ [29]	0.8960	0.7937	0.7545	0.9116	0.9150	0.0150	0.8827	0.8184	0.7970	0.9139	0.9083	0.0220	0.8094	0.7701	0.7554	0.8687	0.8628	0.0569
DNTD$_{22}$ [30]	0.8957	0.7962	0.7288	0.9225	0.9047	0.0113	0.8698	0.8020	0.7645	0.9086	0.9081	0.0217	0.8444	0.8208	0.8065	0.9050	0.8963	0.0425
ICON$_{23}$ [31]	0.9185	0.8371	0.8065	0.9619	0.9497	0.0073	0.9256	0.8671	0.8444	0.9637	0.9554	0.0116	0.8752	0.8664	0.8531	0.9438	0.9239	0.0282
PICR-Net$_{23}$ [32]	0.9307	0.8626	0.8354	0.9628	0.9584	0.0063	0.9463	0.9060	0.8950	0.9744	0.9695	0.0073	0.8859	0.8811	0.8790	0.9445	0.9298	0.0278
LAFB$_{24}$ [33]	0.9325	0.8689	0.8437	0.9690	0.9620	0.0066	0.9462	0.9039	0.8862	0.9798	0.9720	0.0085	0.8763	0.8713	0.8687	0.9412	0.9245	0.0281
HENet$_{25}$ [34]	0.9306	0.8638	0.8319	0.9728	0.9580	0.0064	0.9422	0.9061	0.8938	0.9776	0.9741	0.0082	0.8852	0.8797	0.8727	0.9485	0.9269	0.0252
ORSI SOD Methods																		
MCCNet$_{22}$ [35]	0.9327	0.8604	0.8137	0.9685	0.9538	0.0066	0.9437	0.9054	0.8957	0.9758	0.9735	0.0087	0.8746	0.8630	0.8592	0.9348	0.9182	0.0316
HFANet$_{22}$ [36]	0.9380	0.8681	0.8365	0.9679	0.9644	0.0070	0.9399	0.8981	0.8819	0.9712	0.9722	0.0092	0.8767	0.8624	0.8323	0.9336	0.9191	0.0314
SeaNet$_{23}$ [20]	0.9208	0.8519	0.8304	0.9651	0.9602	0.0073	0.9260	0.8772	0.8625	0.9722	0.9670	0.0105	0.8722	0.8591	0.8556	0.9363	0.9197	0.0308
ESGNet$_{23}$ [37]	0.9373	0.8752	0.8632	0.9733	0.9733	0.0051	0.9484	0.9113	0.9076	0.9815	0.9826	0.0065	0.8795	0.8733	0.8740	0.9438	0.9268	0.0289
GeleNet$_{23}$ [38]	0.9376	0.8781	0.8641	0.9766	0.9750	0.0064	0.9469	0.9128	0.9035	0.9815	0.9786	0.0079	0.8862	0.8785	0.8755	0.9478	0.9265	0.0264
SFANet$_{24}$ [17]	0.9349	0.8680	0.8492	0.9726	0.9669	0.0058	0.9453	0.9063	0.8984	0.9789	0.9765	0.0077	0.8761	0.8659	0.8647	0.9385	0.9202	0.0292
MCP-Net$_{24}$ [39]	0.9373	0.8620	0.8077	0.9657	0.9515	0.0070	0.9433	0.8979	0.8726	0.9736	0.9724	0.0090	0.8736	0.8563	0.8458	0.9331	0.9359	0.0324
ASNet$_{24}$ [18]	0.9345	0.8770	0.8672	0.9745	0.9748	0.0055	0.9441	0.9072	0.8986	0.9764	0.9795	0.0081	-	-	-	-	-	-
WeightNet$_{24}$ [19]	0.9427	0.8800	0.8437	0.9760	0.9674	0.0053	0.9536	0.9159	0.9043	0.9826	0.9801	0.0068	0.8786	0.8635	0.8613	0.9333	0.9380	0.0297
LiteSalNet$_{25}$ [40]	0.9273	0.8809	0.8796	0.9634	0.9729	0.0063	0.9372	0.9061	0.9056	0.9700	0.9752	0.0090	-	-	-	-	-	-
LGIPNet$_{25}$ [41]	0.9396	0.8781	0.8628	0.9753	0.9719	0.0053	0.9507	0.9129	0.9027	0.9829	0.9809	0.0066	0.8854	0.8668	0.8268	0.9432	0.9213	0.0276
RAMENet$_{25}$ [42]	0.9419	0.8840	0.8678	0.9778	0.9735	0.0050	0.9489	0.9147	0.9097	0.9819	0.9797	0.0070	-	-	-	-	-	-
MRBINet$_{25}$ [43]	0.9351	0.8768	0.8626	0.9721	0.9729	0.0056	0.9474	0.9110	0.9058	0.9812	0.9793	0.0069	0.8824	0.8755	0.8755	0.9425	0.9220	0.0268
SggNet$_{25}$ [44]	0.9279	0.8596	0.8386	0.9689	0.9678	0.0068	0.9342	0.8896	0.8884	0.9695	0.9720	0.0111	-	-	-	-	-	-
BCAR-Net$_{25}$ [45]	0.9361	0.8736	0.8740	0.9728	0.9734	0.0054	0.9465	0.9070	0.9073	0.9799	0.9796	0.0071	0.8757	0.8651	0.8666	0.9365	0.9190	0.0306
DPU-Former$_{25}$ [46]	0.9401	0.8757	0.8461	0.9749	0.9652	0.0056	0.9412	0.9142	0.9024	0.9828	0.9802	0.0062	0.8833	0.8831	0.8816	0.9490	0.9287	0.0263
Ours	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**	**0.9551**	**0.9225**	**0.9225**	**0.9889**	**0.9880**	**0.0055**	**0.8845**	**0.8871**	**0.8889**	**0.9528**	**0.9298**	**0.0230**

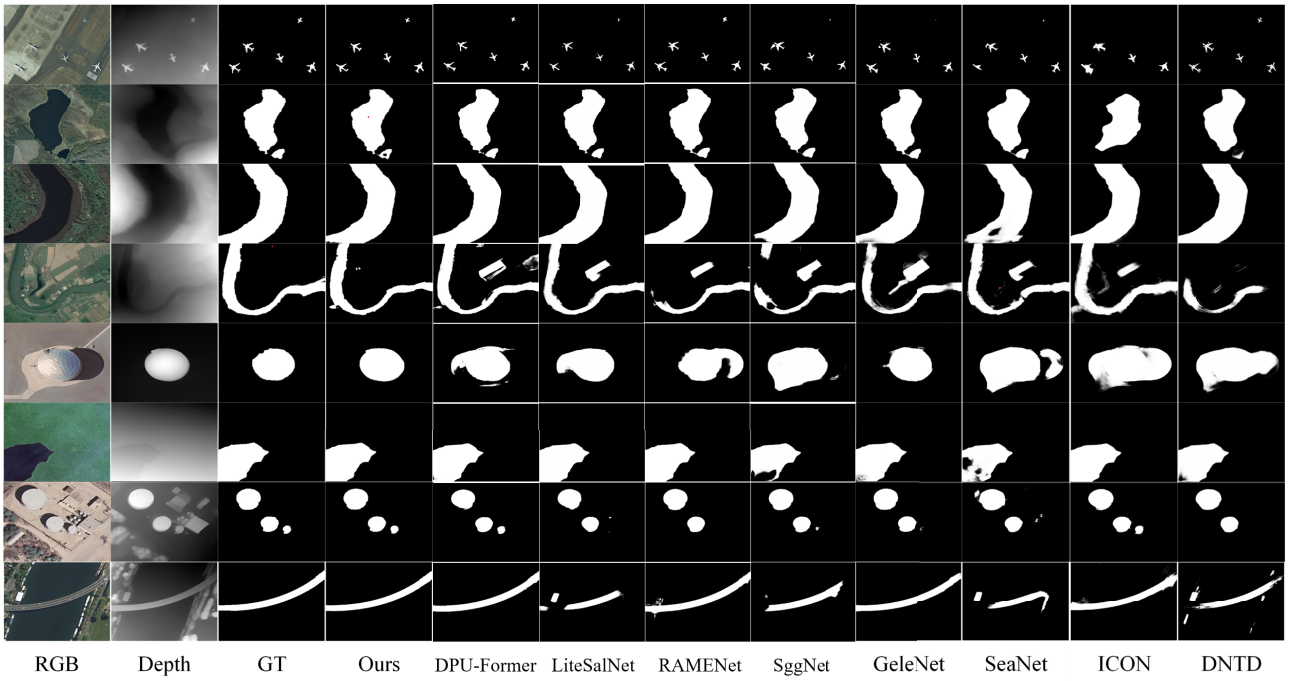


Fig. 6. Visual comparison with eight state-of-the-art methods in some representative ORSI scenarios.

objects, overcoming omissions. In the scenes with large salient objects in Rows 3 and 8, DINet accurately identifies the entire salient object. In the complex contour scenarios of Rows 2 and 6, DINet effectively captures the complete contours of the

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TABLE II

ABLATION EXPERIMENT OF DIFFERENT MODULES. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	En	Fuse	De	EORSSD					
				$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
1	✓			0.9249	0.8615	0.8489	0.9632	0.9666	0.0058
2	✓	✓		0.9315	0.8657	0.8612	0.9710	0.9708	0.0055
3	✓		✓	0.9398	0.8796	0.8770	0.9754	0.9763	0.0052
4	✓	✓	✓	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**

TABLE III

ABLATION EXPERIMENT OF BACKBONES AND ARCHITECTURES. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	Backbone	EORSSD							
		$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$	Params	FLOPs
1	**SMT-share**	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**	**11.98M**	**29.35G**
2	SMT-dual	0.9412	0.8805	0.8779	0.9780	0.9785	0.0044	22.97M	29.35G
3	PVTv2	0.9415	0.8843	0.8728	0.9794	0.9766	0.0041	23.04M	36.51G
4	Res2Net	0.9214	0.8526	0.8483	0.9560	0.9606	0.0070	26.05M	43.17G

objects due to its superior iterative refinement. In addition, in the complex background of Row 4 and the shadow interference in Row 5, DINet benefits from depth cues to more accurately locate the salient objects. The visual results substantiate the effectiveness of DINet across various scenarios.

C. Ablation Experiments

To thoroughly evaluate the contributions and effectiveness of each component of the network, we conducted a series of ablation experiments on the EORSSD dataset. To ensure fair comparisons, all experiments utilized the same training strategy and hyperparameter settings.

1) Ablation Experiments on Module Contributions: The results presented in Table II indicate that both the addition of the fusion module and the decoder to the baseline resulted in varying degrees of performance improvements. For example, incorporating the fusion module led to improvements of 0.66%, 0.42%, and 0.78% for metrics S_α, F_β^m, and E_ξ^m, respectively. In contrast, the inclusion of the decoder resulted in performance enhancements of 1.49%, 1.81%, and 1.22% for the same metrics. This demonstrates that the decoder significantly influences performance, mainly due to the depth cues we introduce, which provide spatial location information while also introducing noise interference. Thus, refining cross-modal features becomes even more crucial. Moreover, the simultaneous addition of both the fusion module and decoder yielded further enhancements of 1.81%, 2.45%, and 1.59% for metrics S_α, F_β^m, and E_ξ^m. These experiments validate the soundness of the network design and the effectiveness of its components.

2) Ablation Experiments on Different Backbones and Architectures: To assess the impact of different backbones on the performance and efficiency of DINet, we conducted the following ablation experiments: 1) shared weight SMT-T [10]; 2) dual-stream SMT-T; 3) PVTv2-b2-li [11]; and 4) Res2Net-50 [47]. The experimental results are summarized in Table III. The comparison between 1) and 2) shows that the shared weight backbone slightly outperforms the dual-stream backbone in performance, benefiting from the initial interaction of cross-modal features, while also reducing the parameter count by 47.8%. Thus, we adopted the shared weight backbone strategy in this study. When comparing the performances of different backbones 1), 3), and 4), it is evident that the performance of the Transformer-based backbones 1) and 3) is comparable and significantly superior to that of the CNN-based backbone 4). This indicates that the global modeling capability provided by Transformers is essential for accurately locating remote sensing objects. Furthermore, SMT, utilizing the CNN-Transformer hybrid architecture, achieves reductions in parameter count and computational load of 48% and 19.6%, respectively. Therefore, we selected SMT as the feature extraction backbone for DINet.

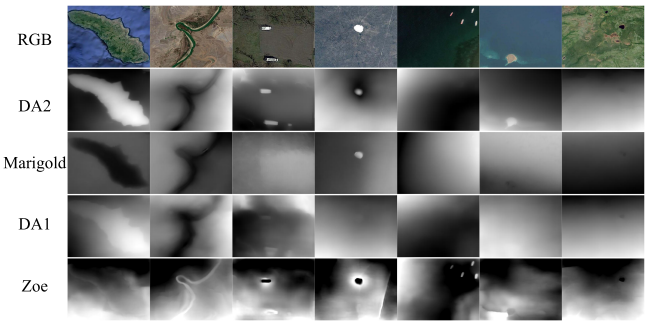


Fig. 7. Visual comparison of depth maps generated by advanced MDE methods.

3) Ablation Experiments on Different Quality Depth Maps: To verify whether the quality of depth maps affects DINet, we employed four advanced MDE methods: Marigold [21], DA [23], DA2 [24], and Zoe [22] to generate depth maps, and then retrained and tested DINet. The generated depth maps and corresponding test results are as follows.

From the results in Fig. 7, it is evident that DA2 achieves relatively excellent depth estimation results for most scenes while producing minimal noise. Zoe also provides depth estimation results for many scenes, but introduces a significant amount of noise. The depth estimation results from Marigold and DA exhibit less noise but show lower prediction success rates. Notably, due to the complexity and specificity of ORSI, even the best-performing method, DA2, encounters some situations where accurate depth estimation is not possible, consequently introducing certain negative information. This places higher demands on the feature aggregation capabilities and noise removal abilities of the multimodal ORSI-SOD model.

The results presented in Table IV show that Zoe and DA experience performance declines compared to the case where depth cues are not utilized, primarily due to the excessive negative information introduced and lower depth estimation success rates, respectively. Marigold shows improvements in metrics F_β^m, F_β^{adp}, E_ξ^{adp}, and $\mathcal{M}$, although it experiences slight declines in metrics S_α and E_ξ^m. DA2 demonstrates significant enhancements across all metrics. In summary, the quality of depth maps is crucial for multimodal ORSI-SOD, and higher

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TABLE IV

ABLATION EXPERIMENT OF DEPTH MAPS. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	Variants	EORSSD					
		$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
1	w/o Depth	0.9424	0.8839	0.8816	0.9783	0.9798	0.0046
2	Zoe	0.9416	0.8860	0.8837	0.9786	0.9778	0.0049
3	DA	0.9404	0.8852	0.8832	0.9779	0.9785	0.0048
4	Marigold	0.9421	0.8854	0.8836	0.9781	0.9801	0.0045
5	**DA2**	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**

TABLE V

ABLATION EXPERIMENT OF APE MODULE. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	Variants	EORSSD					
		$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
1	w/o CA	0.9403	0.8825	0.8794	0.9777	0.9784	0.0044
2	w/o SA	0.9427	0.8845	0.8814	0.9794	0.9799	0.0045
3	w/o CA&SA	0.9398	0.8796	0.8770	0.9754	0.9763	0.0052
4	SA-CA	0.9396	0.8789	0.8758	0.9790	0.9794	0.0048
5	CBAM	0.9411	0.8835	0.8793	0.9782	0.9775	0.0046
6	**DINet**	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**

TABLE VI

ABLATION EXPERIMENT OF ITERATIVE GUIDANCE REFINEMENT DECODER. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	Variants	EORSSD					
		$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
1	w/o IGU	0.9409	0.8810	0.8780	0.9787	0.9791	0.0044
2	w/o RAU	0.9402	0.8724	0.8696	0.9769	0.9619	0.0049
3	w/o IGU&RAU	0.9315	0.8657	0.8612	0.9710	0.9708	0.0055
4	**DINet**	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**

TABLE VII

ABLATION EXPERIMENT OF THE NUMBER OF ITERATIONS. THE BEST RESULTS ARE SHOWN IN **RED**

Times	EORSSD					
	$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
0	0.9409	0.8810	0.8780	0.9787	0.9565	0.0048
1	0.9416	0.8838	0.8826	0.9791	0.9762	0.0044
2	0.9424	0.8857	0.8836	0.9790	0.9802	0.0045
3	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**
4	0.9427	0.8847	0.8819	0.9806	0.9791	0.0043

TABLE VIII

ABLATION EXPERIMENT OF LOSS FUNCTION. THE BEST RESULTS ARE SHOWN IN **RED**

NO.	Variants	EORSSD					
		$S_\alpha \uparrow$	$F_\beta^m \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^m \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
1	BCE-multi	0.9309	0.8571	0.8368	0.9590	0.9403	0.0056
2	IOU-multi	0.9401	0.8736	0.8721	0.9759	0.9766	00049
3	BCE+IOU-single	0.9402	0.8774	0.8697	0.9744	0.9661	0.0051
4	**BCE+IOU-multi**	**0.9430**	**0.8860**	**0.8839**	**0.9791**	**0.9802**	**0.0043**

TABLE IX

QUANTITATIVE EXPERIMENTS ON ROBUSTNESS UNDER NOISE INTERFERENCE

Methods	Noise Variance	EORSSD				Decline
		$E_\xi^m \uparrow$	$S_\alpha \uparrow$	$F_\beta^{adp} \uparrow$	$\mathcal{M} \downarrow$	
RAMENet_{25}	0	0.9778	0.9419	0.8678	0.0050	-
	0.01	0.9178	0.8615	0.7819	0.0177	8.19%
	0.05	0.7744	0.7125	0.5411	0.0326	27.61%
	0.1	0.6826	0.6247	0.3593	0.0385	40.82%
DINet	0	0.9791	0.9430	0.8839	0.0043	-
	0.01	0.9650	0.9220	0.8613	0.0075	2.08%
	0.05	0.8985	0.8652	0.7773	0.0151	9.51%
	0.1	0.8170	0.8058	0.6871	0.0198	17.79%

quality depth maps can further improve the performance of DINet. Therefore, we anticipate more research on MDE specifically for ORSI, which will further advance the development of multimodal ORSI-SOD.

4) Ablation Experiments on the APE Module: To verify the effectiveness of the APE module, we conducted the following experiments: 1) removing the attention perception in the channel dimension; 2) removing the attention perception in the spatial dimension; 3) simultaneously removing the attention perception in both the channel and spatial dimensions; 4) altering the sequence of attention perception by performing spatial attention perception first, followed by channel attention perception; 5) replacing the APE module with the CBAM module, which is also based on channel-space attention; and 6) using the APE module employed by DINet.

The experimental results in Table V fully validate the effectiveness and novelty of the APE. The performance drops observed when removing attention mechanisms from either the channel or spatial dimension, or when altering their execution order, demonstrate that our proposed multidimensional cooperative attention design is both necessary and effective. More importantly, a direct replacement of the APE module with other general-purpose attention mechanisms (e.g., CBAM) also leads to a significant performance degradation. This comparison strongly confirms that our proposed APE module, centered on "oriented perception and cross-interaction," is more suitable for the ORSI-SOD task than generic attention mechanisms in effectively aggregating cross-modal information and enhancing feature representation capabilities.

5) Ablation Experiments on the Iterative Guidance Refinement Decoder: To assess the effectiveness of the iterative guidance refinement decoder, we conducted ablation studies on its components, IGU and RAU, as well as the entire decoder. The results are summarized in Table VI. The results show that utilizing either IGU or RAU individually yields performance gains. Furthermore, the collaborative functioning of both IGU and RAU leads to additional performance enhancements for the decoder. In addition, we performed experiments presented in Table VII to validate the rationale behind the chosen number of iterations for IGU. The results indicate that without iteration, the insufficient refinement of features does not adequately filter out negative information, resulting in suboptimal performance. As the number of iterations increases, performance

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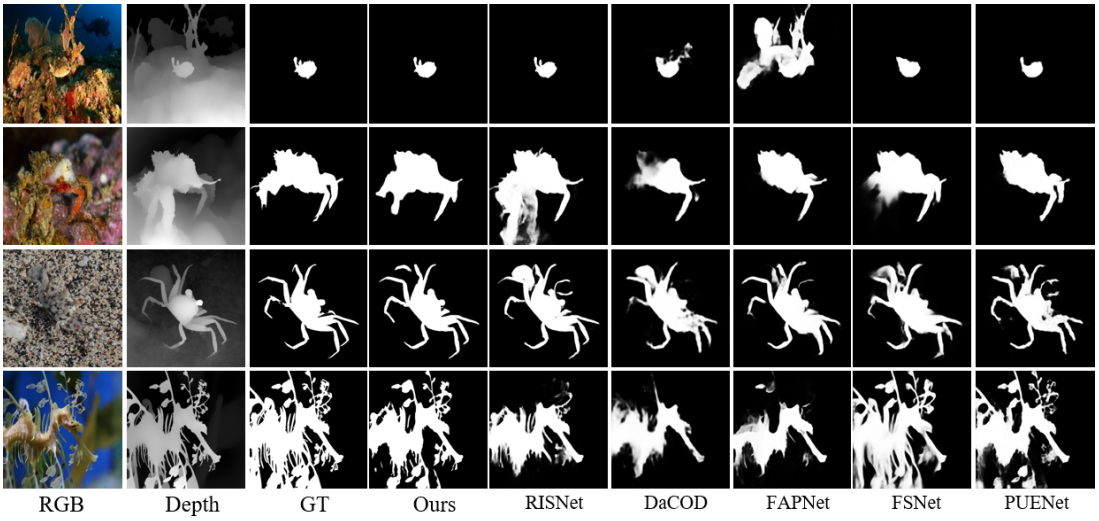


Fig. 8. Visual comparison with state-of-the-art RGB and RGB-D COD methods in some representative COD scenarios.

TABLE X
QUANTITATIVE COMPARISON RESULTS OF DINET AND SIX SOTA COD METHODS ON FOUR BENCHMARK DATASETS, AND THE OPTIMAL RESULTS ARE IN **RED**

Methods	CAMO				CHAMELEON				COD10K				NC4K			
	$E_\xi^m \uparrow$	$S_\alpha \uparrow$	$F_\beta^{adp} \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^m \uparrow$	$S_\alpha \uparrow$	$F_\beta^{adp} \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^m \uparrow$	$S_\alpha \uparrow$	$F_\beta^{adp} \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^m \uparrow$	$S_\alpha \uparrow$	$F_\beta^{adp} \uparrow$	$\mathcal{M} \downarrow$
PUENet_{23} [48]	0.925	0.877	0.856	0.045	0.945	0.910	0.869	0.022	0.926	0.873	0.791	0.022	0.937	0.898	0.866	0.029
FSNet_{23} [49]	0.928	0.880	0.862	0.041	0.951	0.905	0.860	0.023	0.931	0.870	0.793	0.023	0.934	0.891	0.861	0.031
DaCOD_{23} [50]	0.905	0.855	0.831	0.051	0.939	0.893	0.829	0.026	0.907	0.840	0.740	0.028	0.924	0.874	0.833	0.035
RISNet_{24} [51]	0.922	0.870	0.844	0.050	-	-	-	-	0.931	0.873	0.804	0.025	0.925	0.882	0.851	0.037
DSAM_{24} [52]	0.913	0.832	0.821	0.061	0.942	0.883	0.877	0.028	0.921	0.846	0.789	0.033	0.932	0.871	0.847	0.040
MultiCOS_{25} [53]	0.923	0.870	0.845	0.048	0.962	**0.920**	0.884	0.019	0.933	0.874	0.809	0.021	0.932	0.887	0.859	0.032
Ours	**0.935**	**0.884**	**0.877**	**0.036**	**0.974**	0.916	**0.904**	**0.016**	**0.936**	**0.876**	**0.837**	**0.019**	**0.940**	**0.892**	**0.885**	**0.028**

progressively improves and stabilizes after three iterations; thus, we have set the number of iterations to 3.

6) Ablation Experiments on the Loss Function: To validate the effectiveness of the loss function and supervision strategy employed, we conducted the following experiments: 1) supervising multilevel predictions with the BCE loss function; 2) supervising multilevel predictions with the IoU loss function; 3) supervising final predictions using a hybrid loss function composed of both BCE and IoU; and 4) supervising multilevel predictions with a hybrid loss function composed of BCE and IoU. The experimental results are presented in Table VIII. The BCE loss function focuses on the overall pixel distribution, while the IoU loss function emphasizes foreground area information. As a result, both BCE loss and IoU loss, when used individually, concentrate on partial information, leading to suboptimal overall performance. In contrast, the hybrid loss function combines the strengths of both, providing more comprehensive supervision and achieving significant improvements in results. In addition, comparing the results of 3) and 4) demonstrates that supervising the multilevel predictions generated through iterations, alongside the final predictions, effectively utilizes the shared information from multilevel features, enhancing the robustness of the network's discrimination and significantly improving the prediction results.

7) Robustness Analysis: Since real-world application scenarios are often more challenging than public datasets, we conducted the following experiments to evaluate the practical applicability of DINet under complex and adverse settings. Specifically, we introduced varying levels of simulated Gaussian noise (with zero mean and variances of 0.01, 0.05, and 0.1) to the test input images to assess the model's ability to maintain robustness.

As summarized in Table IX, DINet demonstrates significantly stronger robustness compared to RAMENet [42] under all noise levels. For instance, under slight noise (variance = 0.01), DINet exhibits only a minimal performance decline of 2.08% in the composite index, whereas RAMENet suffers a more substantial drop of 8.19%. Under the highest noise level (variance = 0.1), DINet maintains relatively stable performance with a 17.79% decline, while RAMENet experiences a severe performance drop of 40.82%. This superior robustness stems from DINet's deep information interaction and iterative refinement design. These mechanisms enable the model to effectively suppress noise propagation while preserving structural and semantic consistency. Even under relatively severe noise conditions, DINet remains relatively robust, confirming its suitability for real-world applications where input quality may be compromised.

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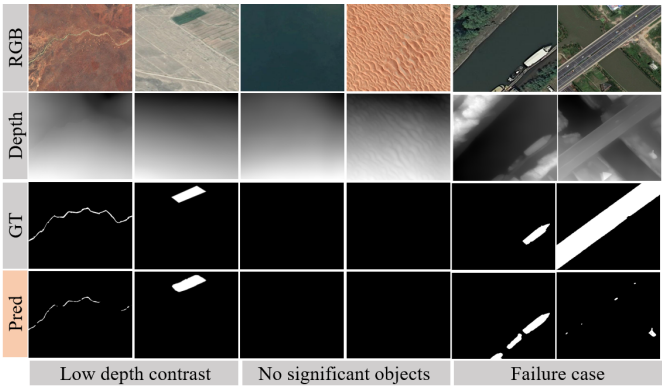


Fig. 9. Visualization results of DINet in extreme scenarios.

D. Application

Camouflaged object detection (COD) aims to locate and segment camouflaged objects hidden within complex scenes, which is a challenging task. Similar to ORSI-SOD, COD also faces the challenge of difficult object localization within complex environments and lacks available depth data. Therefore, to fairly compare DINet with multimodal input methods and explore its applicability in different scenarios, we applied it to the COD task.

Following the setup described in other COD articles, we used 4040 images from CAMO [54] and COD10K [55] as the training set, and then tested on the CAMO and COD10K testing sets, as well as the entire NC4K [56] and CHAMELEON [57] datasets. We compared DINet with RGB COD methods PUENet [48], FSNet [49], and RGB-D COD methods, including DaCOD [50], RISNet [51], DSAM [52], and MultiCOS [53].

The quantitative comparison results are shown in Table X. The proposed DINet achieved the best results in nearly all metrics across the four datasets, with significant improvements over other methods. In addition, as observed in the visual comparisons in Fig. 8, DINet excels in handling representative scenarios, such as small objects and complex contours, thanks to its depth cue-assisted localization and iterative refinement capabilities. These experiments demonstrate that DINet, employing a multimodal strategy, exhibits strong applicability across various scenarios in the COD task.

E. Discussion on Specific Extreme Scenarios

As shown in Fig. 9, to evaluate the robustness and limitations of DINet across different scenarios, we provide visualizations and discussions of several specific complex ORSI scenes. Columns 1 and 2 display scenes with low depth contrast between salient objects and their surroundings; Columns 3 and 4 present scenes containing no salient objects, while Columns 5 and 6 show representative failure cases.

It can be observed that in the first two scenarios, depth maps fail to provide valuable discriminative information, which consequently affects the prediction results. However, DINet mitigates such interference through its iterative refinement mechanism and maintains robustness. In Column 5, ambiguous depth maps introduce erroneous semantic information, leading the model to perceive objects at different depths as continuous structures. In Column 6, uncertainty in salient object categories (where bridges, cars, and houses could all be considered prominent objects in ORSI) results in incorrect identification of the salient target. In summary, DINet remains robust even when depth information is unreliable, yet it exhibits certain limitations in semantic information processing. In future work, we will explore incorporating category information to enhance the model's discriminative capability.

V. Conclusion

In this article, we proposed DINet, a novel DINet for ORSI-SOD. By incorporating depth information as auxiliary cues, DINet effectively addresses inherent ORSI-SOD challenges. These include difficulties in object localization, missed detections, and boundary uncertainties. Through a carefully designed architecture comprising a shared-weight encoder, an APE module, and an iterative guidance refinement decoder, DINet achieves robust and accurate SOD while maintaining lightweight and real-time performance. Extensive experiments validating the effectiveness of the multimodal strategy and the proposed modules. Ablation studies further confirm DINet's robustness under noisy conditions. This suggests significant potential for real-world applications, such as disaster response, urban planning, and environmental monitoring. The ability to accurately locate and segment prominent objects in complex remote sensing scenes can greatly improve the efficiency and reliability of these systems. However, DINet exhibits certain limitations in semantic information processing. In future work, we will incorporate semantic category information to resolve ambiguity, explore ORSI-specific MDE to enhance cue quality, and optimize the multimodal architecture to improve computational efficiency.

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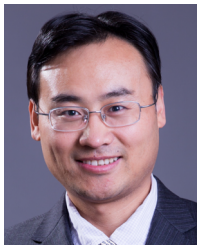
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